

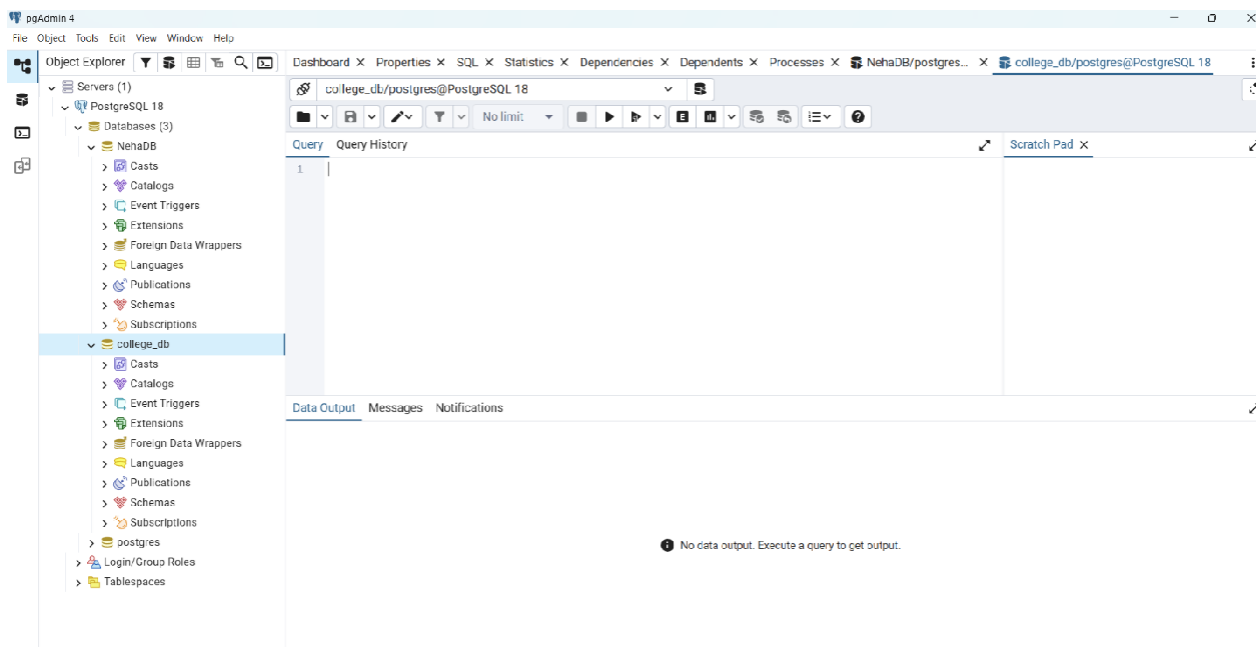
Module 3 :Database Integration HandsOn

NEHA S

HandsOn-1

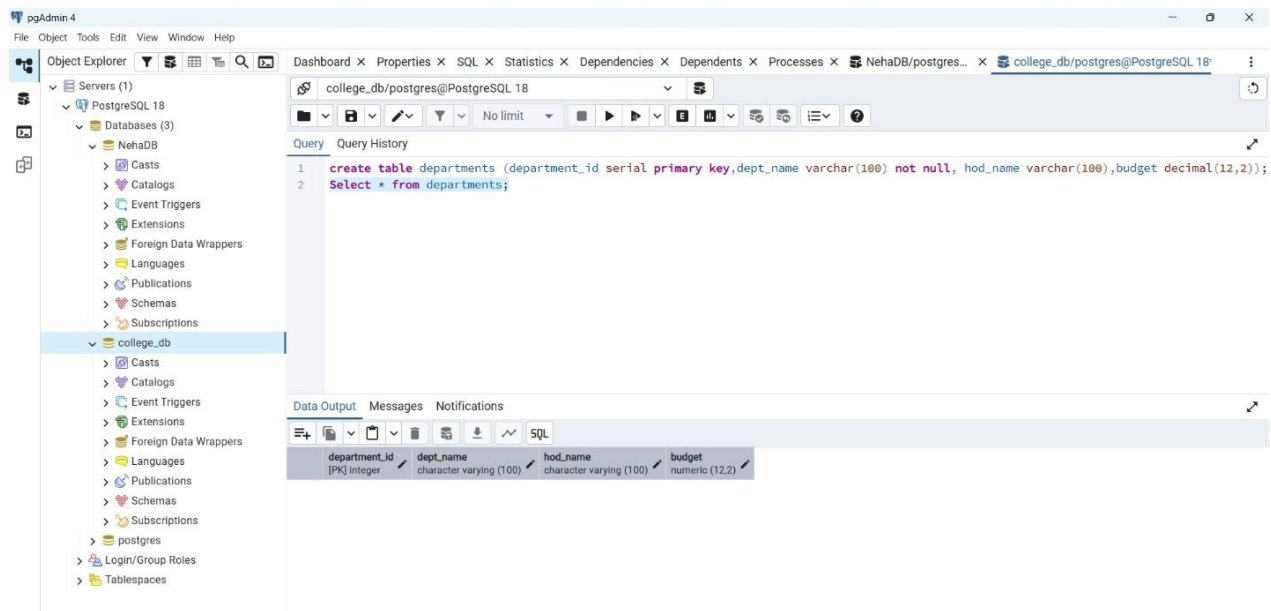
TASK 1: Create the Database and Tables Goal: Practise DDL by N
creating the full college_db schema with constraints

1. Create a new database named college_db.

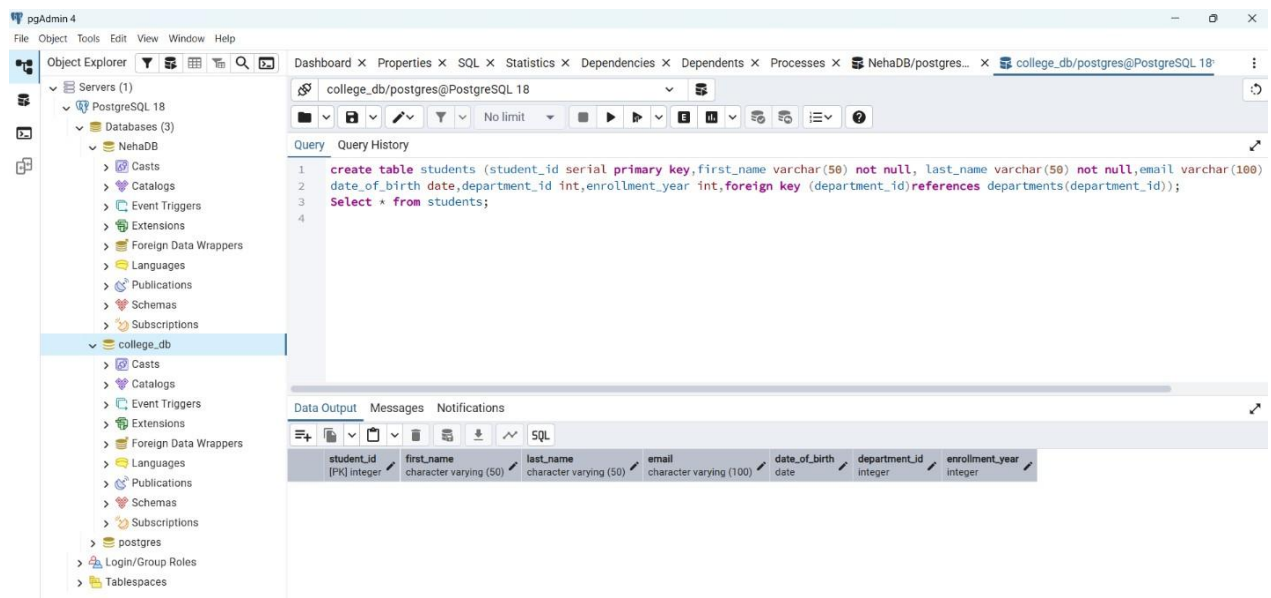


2. Write CREATE TABLE statements for all five tables listed in the schema section:
departments, students, courses, enrollments, professors.

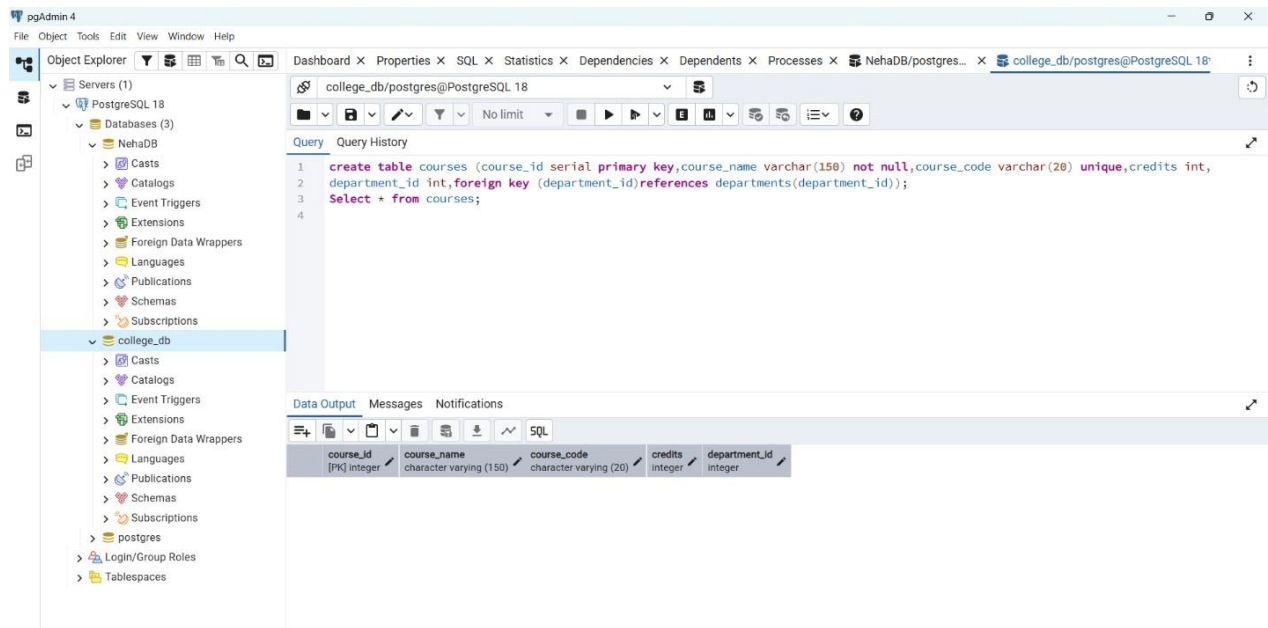
departments:



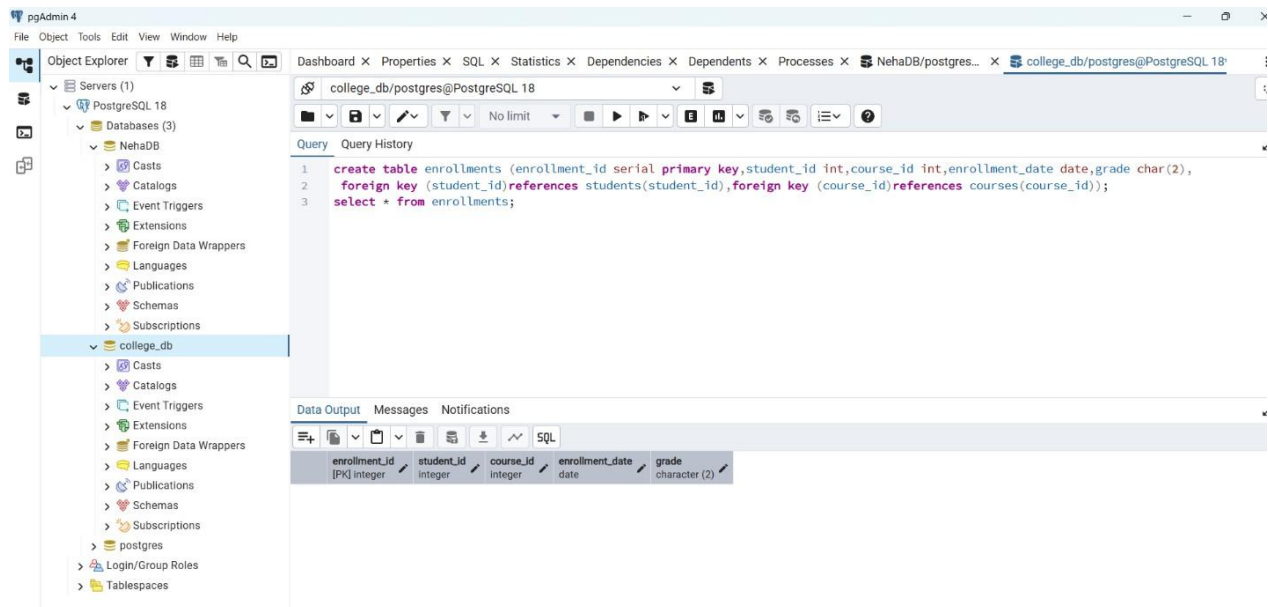
students:



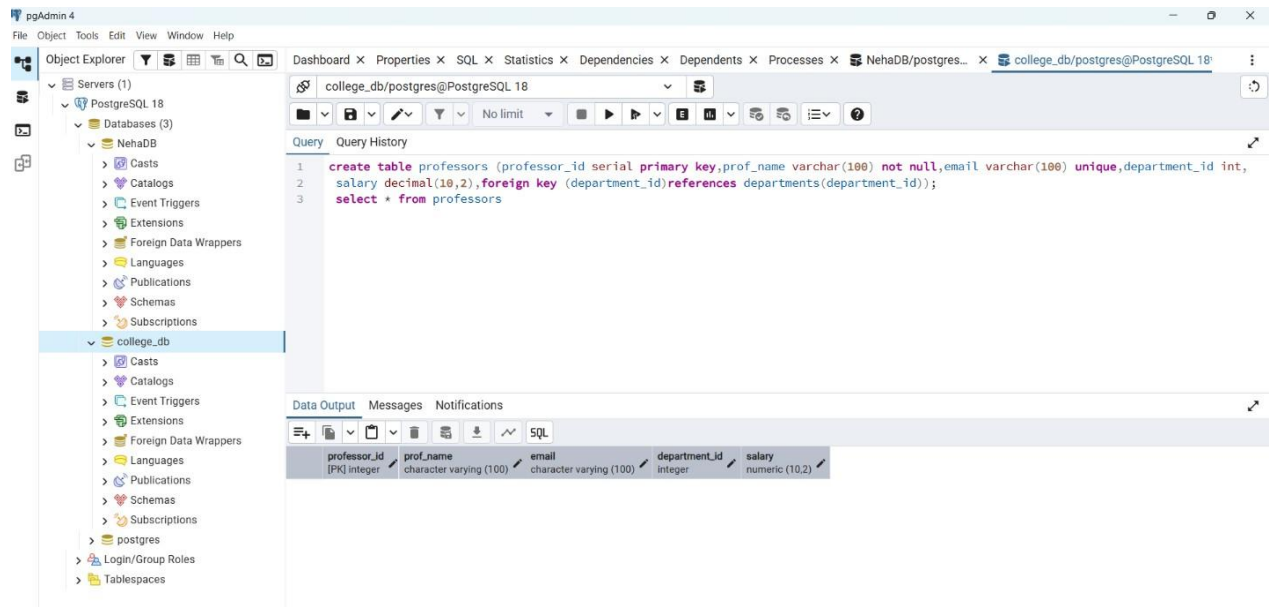
courses:



enrollments:

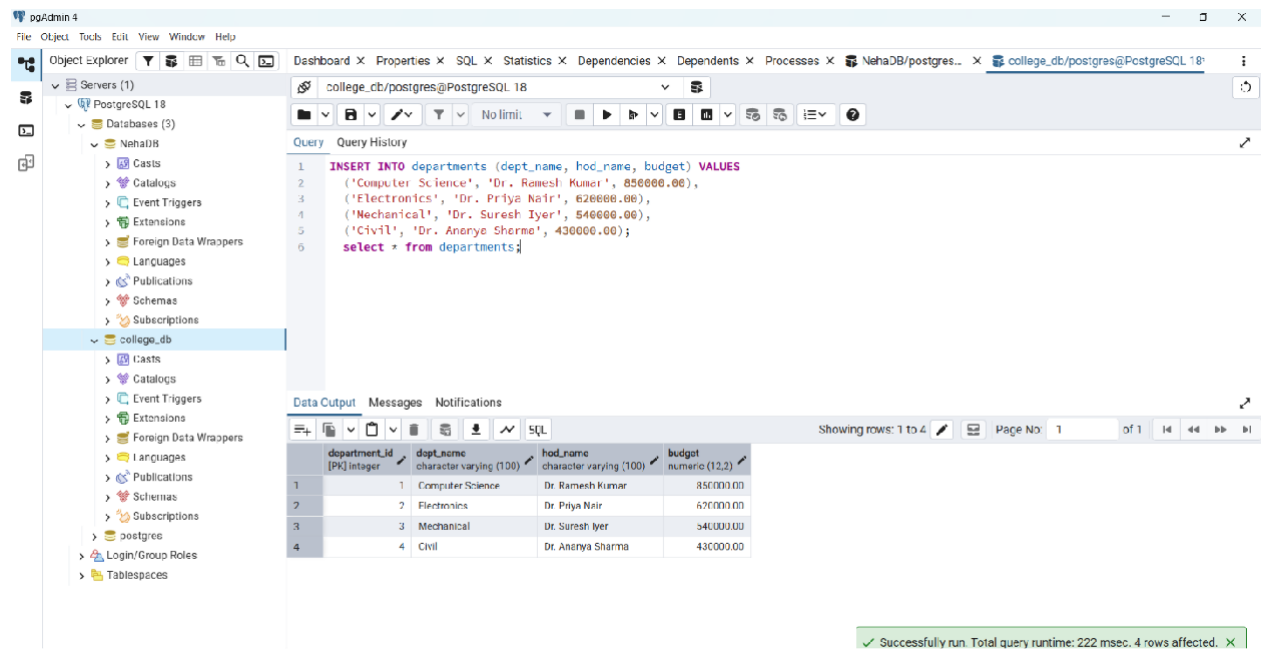


professors:



15. Insert table:

departments:



students:

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

Query Query History

```
1 INSERT INTO students (first_name, last_name, email, date_of_birth, department_id,  
2 enrolment_year) VALUES  
3 ('Arjun', 'Mehta', 'arjun.mehta@college.edu', '2003-04-12', 1, 2022),  
4 ('Priya', 'Suresh', 'priya.suresh@college.edu', '2003-07-25', 1, 2022),  
5 ('Rohan', 'Verma', 'rohan.verma@college.edu', '2002-11-08', 2, 2021),  
6 ('Sneha', 'Patel', 'sneha.patel@college.edu', '2004-01-30', 3, 2023),  
7 ('Vikram', 'Das', 'vikram.das@college.edu', '2003-09-14', 1, 2022),  
8 ('Kavya', 'Menon', 'kavya.menon@college.edu', '2002-05-17', 2, 2021),  
9 ('Aditya', 'Singh', 'aditya.singh@college.edu', '2004-03-22', 4, 2023),  
10 ('Deepika', 'Rao', 'deepika.rao@college.edu', '2003-08-09', 1, 2022);  
11 select * from students;
```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

student_id	first_name	last_name	email	date_of_birth	department_id	enrolment_year
1	Arjun	Mehta	arjun.mehta@college.edu	2003-04-12	1	2022
2	Priya	Suresh	priya.suresh@college.edu	2003-07-25	1	2022
3	Rohan	Verma	rohan.verma@college.edu	2002-11-08	2	2021
4	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	Kavya	Menon	kavya.menon@college.edu	2002-05-17	2	2021
7	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023
8	Deepika	Rao	deepika.rao@college.edu	2003-08-09	1	2022

Courses:

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File Object Tools Edit View Window Help

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

Query Query History

```
1 INSERT INTO courses (course_name, course_code, credits, department_id) VALUES  
2 ('Data Structures & Algorithms', 'CS101', 4, 1),  
3 ('Database Management Systems', 'CS102', 3, 1),  
4 ('Object Oriented Programming', 'CS103', 4, 1),  
5 ('Circuit Theory', 'EC101', 3, 2),  
6 ('Thermodynamics', 'ME101', 3, 3);  
7 select * from courses;
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

course_id	course_name	course_code	credits	department_id
1	Data Structures & Algorithms	CS101	4	1
2	Database Management Systems	CS102	3	1
3	Object Oriented Programming	CS103	4	1
4	Circuit Theory	EC101	3	2
5	Thermodynamics	ME101	3	3

Enrollments:

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File Object Tools Edit View Window Help

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

Query Query History

```

1 INSERT INTO enrollments (student_id, course_id, enrollment_date, grade) VALUES
2 (1, 1, '2022-07-01', 'A'), (1, 2, '2022-07-01', 'B'),
3 (2, 1, '2022-07-01', 'B'), (2, 3, '2022-07-01', 'A'),
4 (3, 4, '2022-07-01', 'A'), (4, 5, '2023-07-01', NULL),
5 (5, 1, '2022-07-01', 'C'), (5, 2, '2022-07-01', 'A'),
6 (6, 4, '2022-07-01', 'B'), (7, 5, '2023-07-01', NULL),
7 (8, 1, '2022-07-01', 'A'), (8, 3, '2022-07-01', 'B')
8 select * from enrollments;

```

Data Output Messages Notifications

Showing rows: 1 to 12 Page No: 1 of 1

enrollment_id	student_id	course_id	enrollment_date	grade
1	1	1	2022-07-01	A
2	2	1	2022-07-01	B
3	3	2	2022-07-01	B
4	4	2	2022-07-01	A
5	5	3	2021-07-01	A
6	6	4	2023-07-01	[null]
7	7	5	2022-07-01	C
8	8	5	2022-07-01	A
9	9	6	2021-07-01	B

Total rows: 12 Query complete: 00:00:00.182 CRLF Ln:8, Col:1

professors:

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

Query Query History

```

1 INSERT INTO professors (prof_name, email, department_id, salary) VALUES
2 ('Dr. Anand Krishnan', 'anand.k@college.edu', 1, 95000.00),
3 ('Dr. Meena Pillai', 'meena.p@college.edu', 1, 88000.00),
4 ('Dr. Sunil Rajan', 'sunil.r@college.edu', 2, 82000.00),
5 ('Dr. Latha Gopal', 'latha.g@college.edu', 3, 79000.00),
6 ('Dr. Kartik Bose', 'kartik.b@college.edu', 4, 76000.00);
7 select * from professors;

```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

professor_id	prof_name	email	department_id	salary
1	Dr. Anand Krishnan	anand.k@college.edu	1	95000.00
2	Dr. Meena Pillai	meena.p@college.edu	1	88000.00
3	Dr. Sunil Rajan	sunil.r@college.edu	2	82000.00
4	Dr. Latha Gopal	latha.g@college.edu	3	79000.00
5	Dr. Kartik Bose	kartik.b@college.edu	4	76000.00

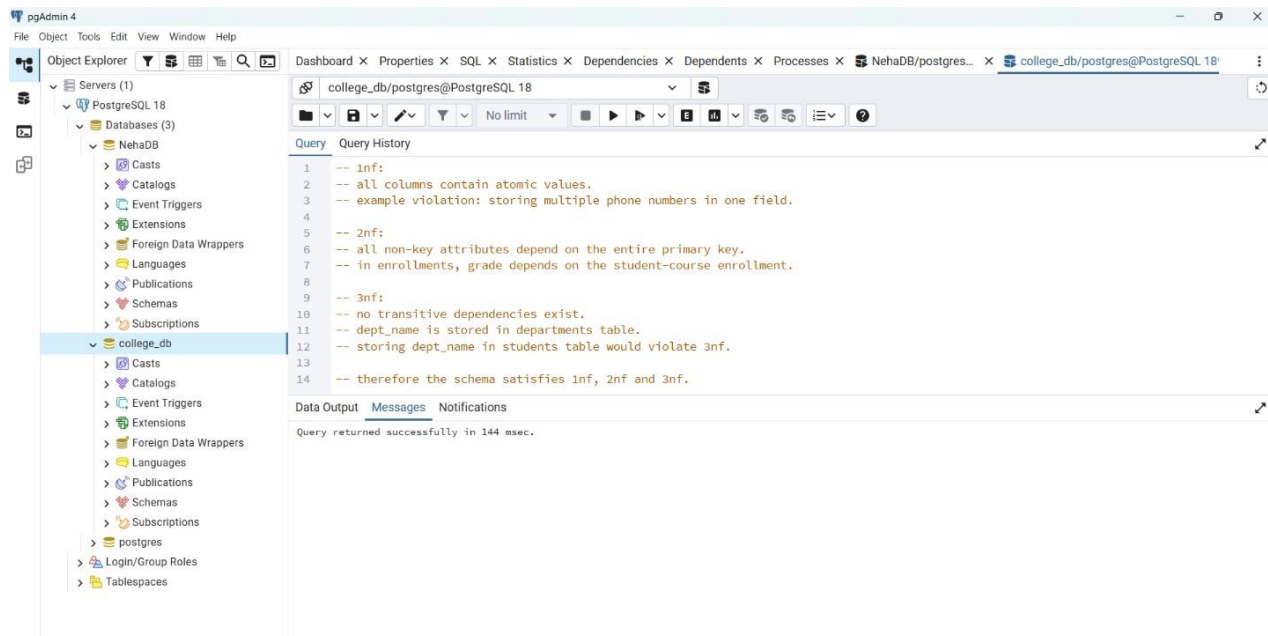
TASK 2:

6. For 1NF: Confirm every column holds atomic values. Identify any hypothetical violation if, for example, we had stored multiple phone numbers in one field.

7. For 2NF: Confirm every non-key column is fully dependent on the entire primary key (especially check the enrollments table, which has a composite candidate key of student_id + course_id).

8. For 3NF: Confirm there are no transitive dependencies. (Hint: would storing dept_name in the students table violate 3NF?)

9. Write a short comment using SQL -- comments inside your .sql file documenting your 3NF analysis for the enrollments table



TASK 3:

10. Add a column phone_number VARCHAR(15) to the students table

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

Query Query History

```

1 alter table students
2 add column phone_number varchar(15);
3 select * from students;

```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

student_id [PK] integer	first_name character varying (50)	last_name character varying (50)	email character varying (100)	date_of_birth date	department_id integer	enrollment_year integer	phone_number character varying (15)
1	Arjun	Mehta	arjun.mehta@college.e...	2003-04-12	1	2022	[null]
2	Priya	Suresh	priya.suresh@college.e...	2003-07-25	1	2022	[null]
3	Rohan	Verma	rohan.verma@college.e...	2002-11-08	2	2021	[null]
4	Sneha	Patel	sneha.patel@college.e...	2004-01-30	3	2023	[null]
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022	[null]
6	Kavya	Menon	kavya.menon@college...	2002-05-17	2	2021	[null]
7	Aditya	Singh	aditya.singh@college.e...	2004-03-22	4	2023	[null]
8	Deepika	Rao	deepika.rao@college.e...	2003-08-09	1	2022	[null]

11. Add a column max_seats INT DEFAULT 60 to the courses table.

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

Query Query History

```

1 alter table courses
2 add column max_seats int default 60;
3 select * from courses;

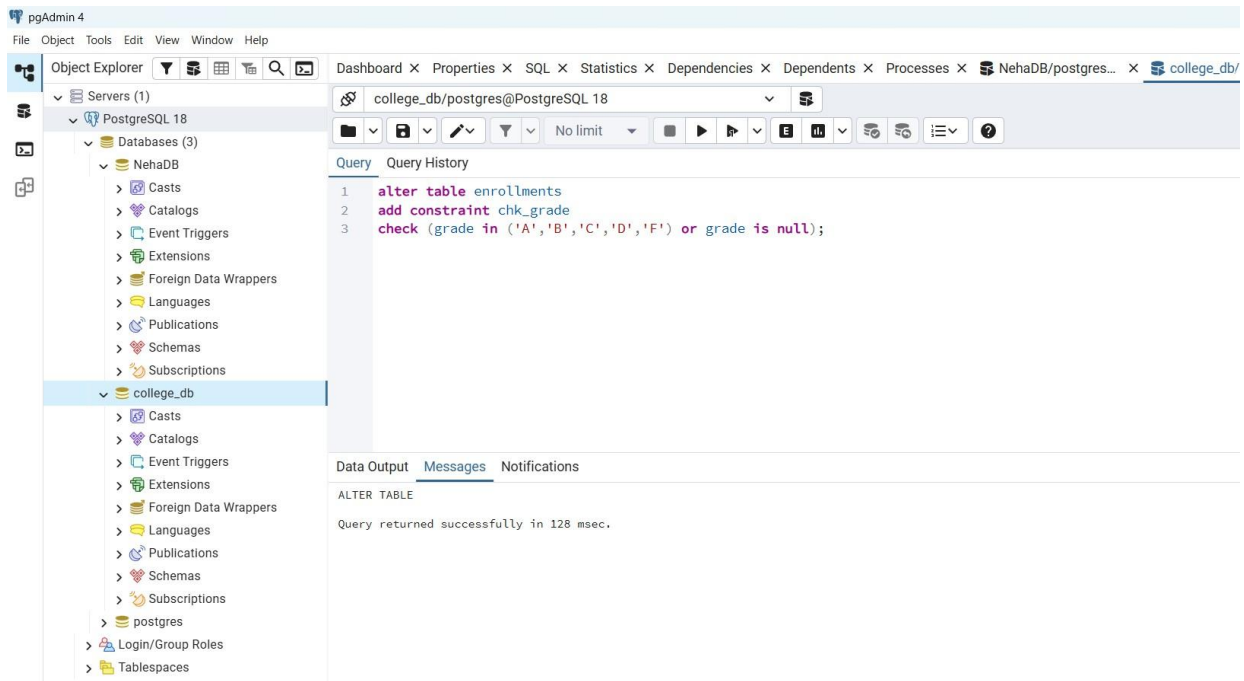
```

Data Output Messages Notifications

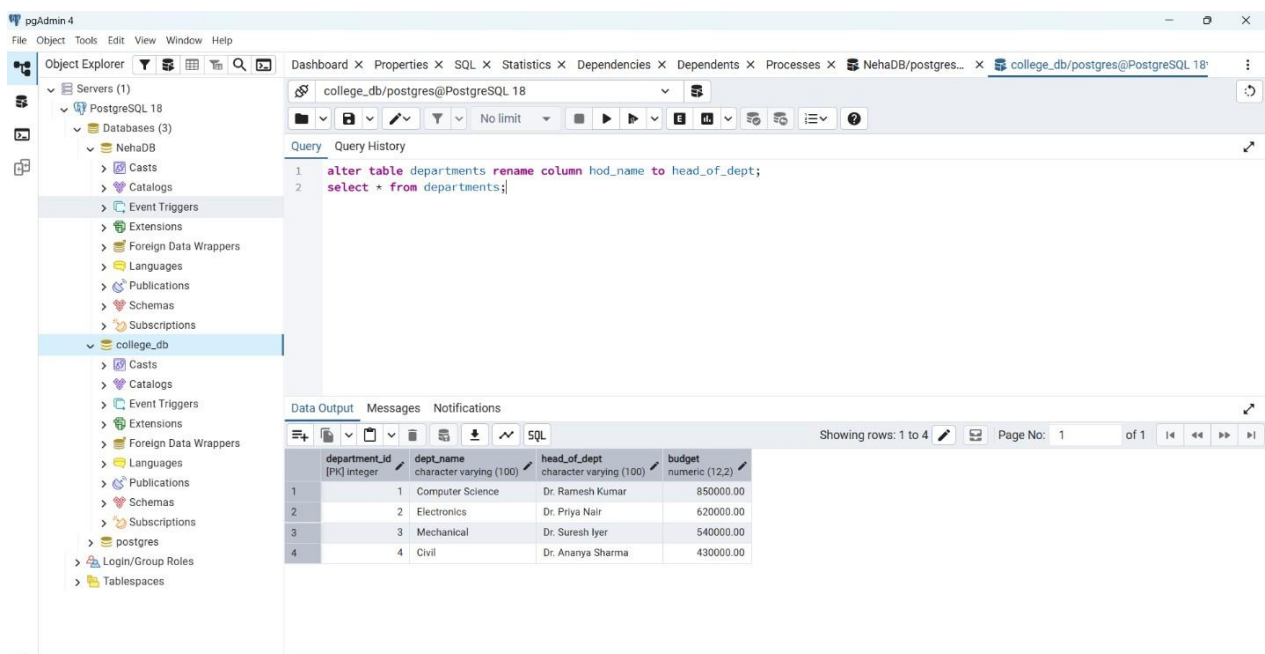
Showing rows: 1 to 5 Page No: 1 of 1

course_id [PK] integer	course_name character varying (150)	course_code character varying (20)	credits integer	department_id integer	max_seats integer
1	Data Structures & Algorith...	CS101	4	1	60
2	Database Management Sy...	CS102	4	1	60
3	Object Oriented Programm...	CS103	3	1	60
4	Circuit Theory	EC101	3	2	60
5	Thermodynamics	ME101	3	3	60

12. Add a CHECK constraint to enrollments ensuring grade is one of ('A','B','C','D','F') or NULL



13. Rename the hod_name column in departments to head_of_dept (PostgreSQL: ALTER COLUMN ... RENAME TO; MySQL: ALTER TABLE ... CHANGE ...).



14. Drop the phone_number column you added in step 1 (simulate a schema rollback)

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

No limit

Query Query History

```

1 alter table students drop column phone_number;
2 select * from students;

```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

	student_id [PK] integer	first_name character varying (50)	last_name character varying (50)	email character varying (100)	date_of_birth date	department_id integer	enrollment_year integer
1	1	Arjun	Mehta	arjun.mehta@college.e...	2003-04-12	1	2022
2	2	Priya	Suresh	priya.suresh@college.e...	2003-07-25	1	2022
3	3	Rohan	Verma	rohan.verma@college.e...	2002-11-08	2	2021
4	4	Sneha	Patel	sneha.patel@college.e...	2004-01-30	3	2023
5	5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	6	Kavya	Menon	kavya.menon@college...	2002-05-17	2	2021
7	7	Aditya	Singh	aditya.singh@college.e...	2004-03-22	4	2023
8	8	Deepika	Rao	deepika.rao@college.e...	2003-08-09	2	2022

Total rows: 8 Query complete 00:00:00.191

Successfully run. Total query runtime: 191 msec. 8 rows

HandsOn-2

TASK 1:

16. Insert two additional students of your own choosing into the students table.

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (3) > NehaDB > college_db

Query:

```

1 insert into students
2 (first_name,last_name,email,date_of_birch,department_id,enrollment_year)
3 values
4 ('rahul','sharma','rahul.sharma@college.edu','2003-05-10',1,2022);
5 ('anita','reddy','anita.reddy@college.edu','2004-02-15',2,2023);
6 select * from students;

```

Data Output:

student_id	first_name	last_name	email	date_of_birth	department_id	enrollment_year
1	Priya	Jaiswal	priya.jaiswal@college.edu	2003-07-20	1	2024
3	Rohan	Verma	rohan.verma@college.edu	2002-11-08	2	2021
4	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	Kavya	Menon	kavya.menon@college.edu	2002-05-17	2	2021
7	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023
8	Deepika	Rao	deepika.rao@college.edu	2003-08-09	1	2022
9	rahul	sharma	rahul.sharma@college.edu	2003-05-10	1	2022
10	anita	reddy	anita.reddy@college.edu	2004-02-15	2	2023

17. Update the grade of student_id = 5 for course_id = 1 from 'C' to 'B'

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (3) > NehaDB > college_db

Query:

```

1 select count(*) from students;

```

Data Output:

count
10

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer Servers (1) PostgreSQL 18 Databases (3) NehaDB college_db

college_db/postgres@PostgreSQL 18

Query Query History

```
1 update enrollments set grade='B' where student_id=5 and course_id=1;
2 select * from enrollments where student_id=5 and course_id=1;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

enrollment_id	student_id	course_id	enrollment_date	grade
1	7	5	2022-07-01	B

18. Delete enrollments where grade IS NULL (students who never received a grade)

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer Servers (1) PostgreSQL 18 Databases (3) NehaDB college_db

college_db/postgres@PostgreSQL 18

Query Query History

```
1 select * from enrollments where grade is null;
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

enrollment_id	student_id	course_id	enrollment_date	grade
1	6	4	2023-07-01	[null]
2	10	7	2023-07-01	[null]

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres

college_db/postgres@PostgreSQL 18

No limit

Query Query History

```
1 select * from enrollments where grade is null;
2 delete from enrollments where grade is null;
3 select * from enrollments;
```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1

enrollment_id	student_id	course_id	enrollment_date	grade
(PK) integer	integer	integer	date	character (2)
1	1	1	2022-07-01	A
2	2	1	2022-07-01	B
3	3	2	2022-07-01	B
4	4	2	2022-07-01	A
5	5	3	2021-07-01	A
6	8	5	2022-07-01	A
7	9	6	2021-07-01	B
8	11	8	2022-07-01	A
9	12	8	2022-07-01	B

Total rows: 10 Rows updated: 00:00:00.100

19. Verify row counts using SELECT COUNT(*) after each operation

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

No limit

Query Query History

```
1 select count(*) from students;
2 select count(*) from enrollments;
```

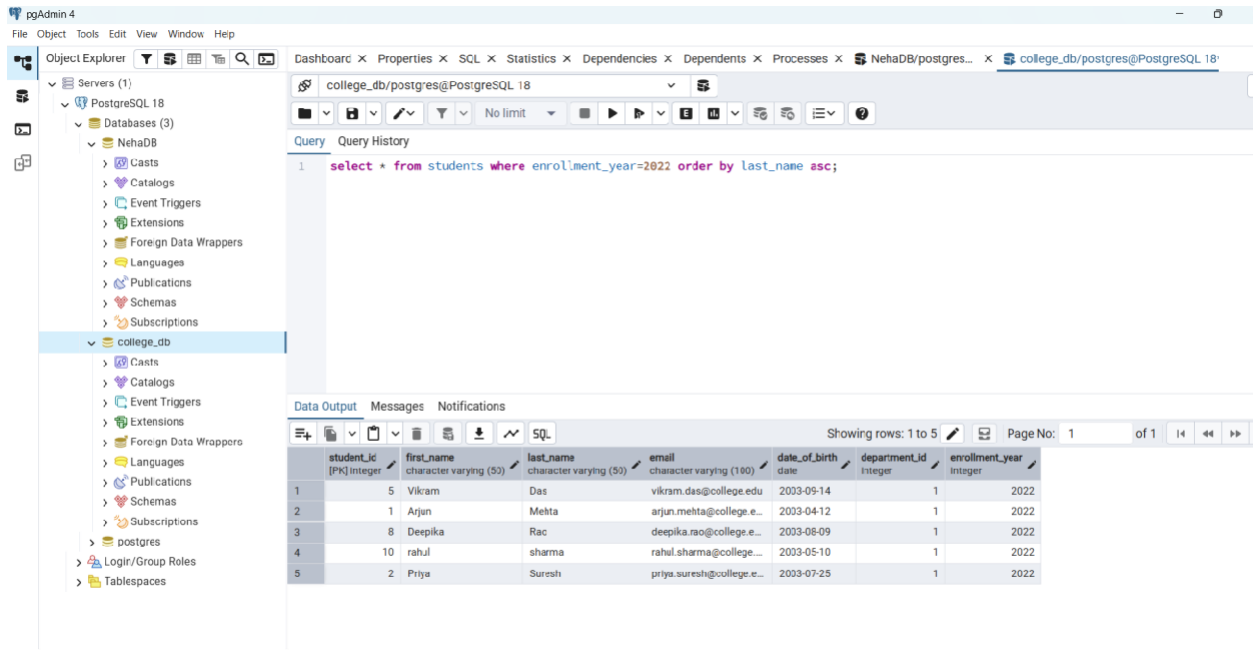
Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

count
bigint
10

TASK 2: Single-Table Queries and Filtering

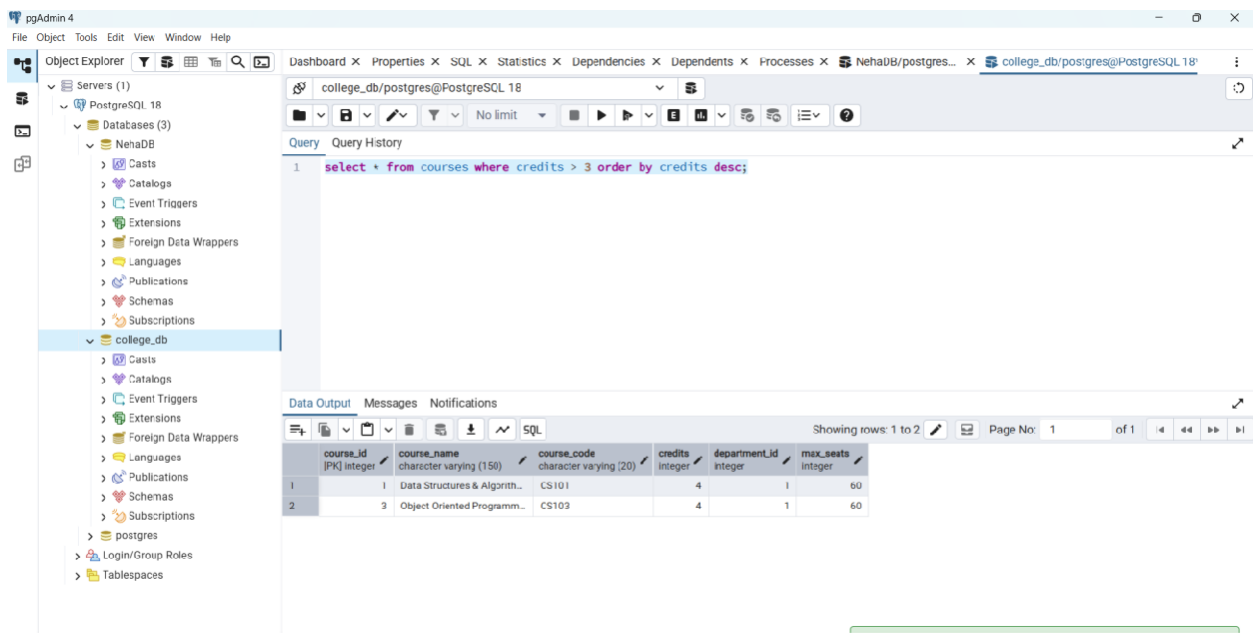
20. Retrieve all students enrolled in 2022, ordered by last_name alphabetically



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query: `select * from students where enrollment_year=2022 order by last_name asc;`. The 'Data Output' tab is active, displaying a table with 5 rows of student data.

student_id [PK] integer	first_name character varying (50)	last_name character varying (50)	email character varying (100)	date_of_birth date	department_id integer	enrollment_year integer
1	5 Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
2	1 Arjun	Melita	arjun.melita@college.e...	2003-04-12	1	2022
3	8 Deepika	Rac	deepika.rao@college.e...	2003-08-09	1	2022
4	10 rahul	sharma	rahul.sharma@college...	2003-05-10	1	2022
5	2 Priya	Suresh	priya.suresh@college.e...	2003-07-25	1	2022

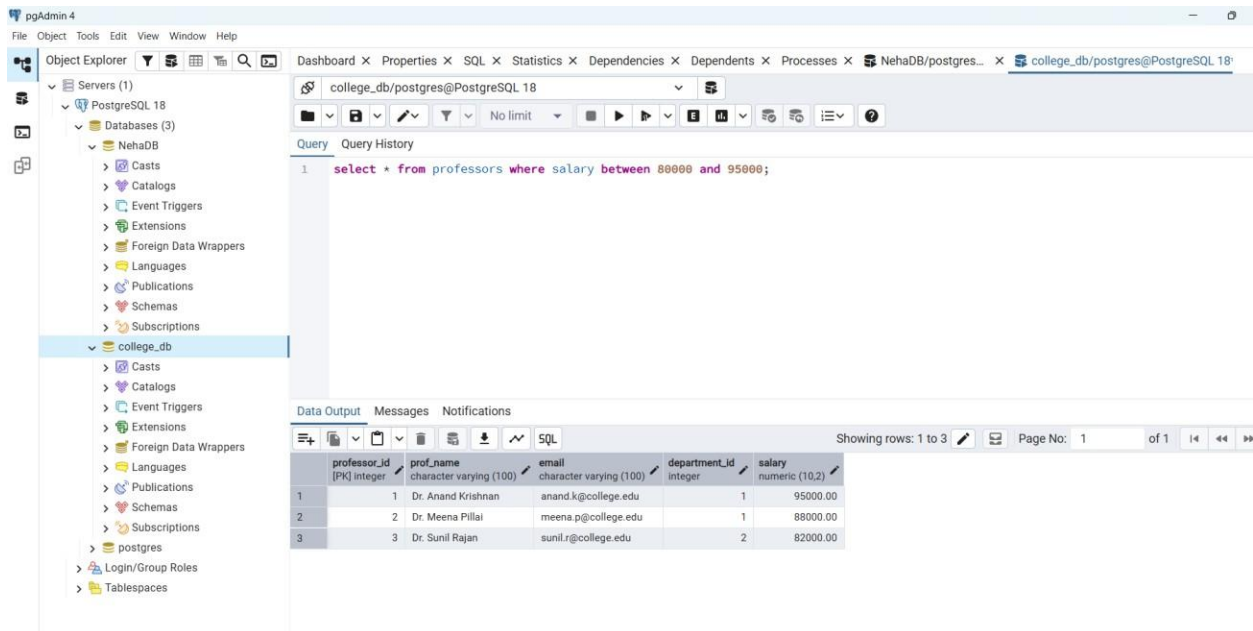
21. Find all courses with more than 3 credits, sorted by credits descending



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query: `select * from courses where credits > 3 order by credits desc;`. The 'Data Output' tab is active, displaying a table with 2 rows of course data.

course_id [PK] integer	course_name character varying (150)	course_code character varying (20)	credits integer	department_id integer	max_seats integer
1	Data Structures & Algorith...	CS101	4	1	60
2	Object Oriented Programmm...	CS102	4	1	60

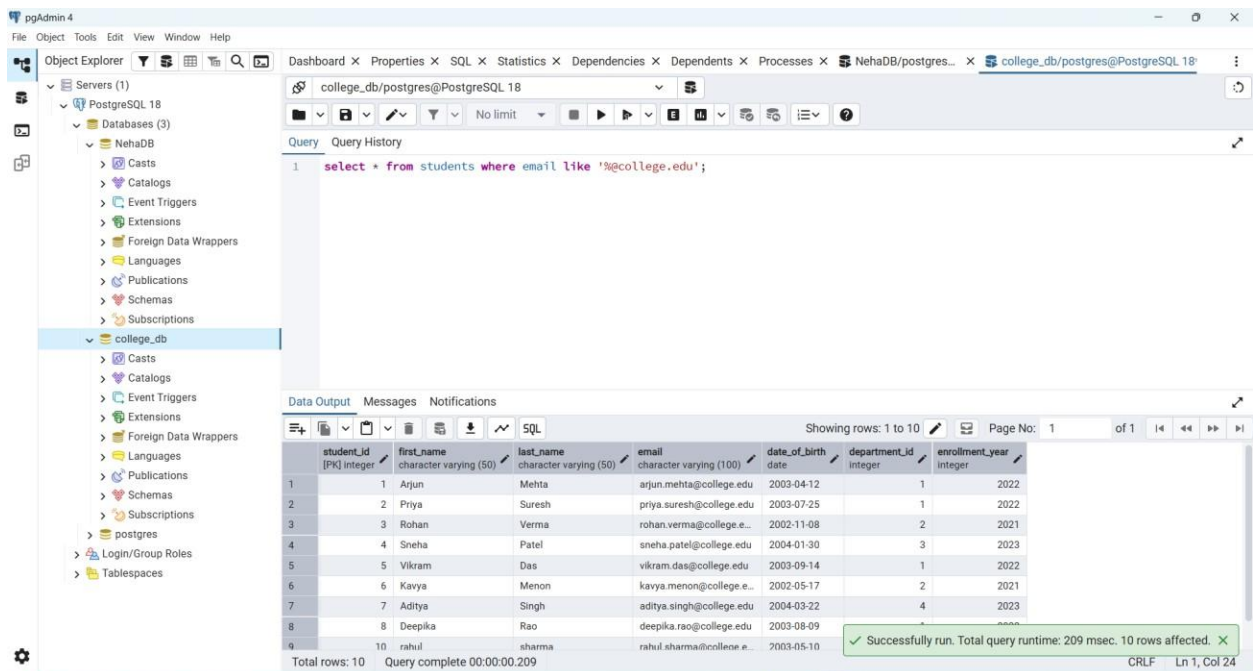
22. List all professors whose salary is between 80,000 and 95,000_.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query: `select * from professors where salary between 80000 and 95000;`. The 'Data Output' tab is active, displaying a table with 3 rows and 5 columns: professor_id, prof_name, email, department_id, and salary.

professor_id [PK] integer	prof_name character varying (100)	email character varying (100)	department_id integer	salary numeric (10,2)
1	Dr. Anand Krishnan	anand.k@college.edu	1	95000.00
2	Dr. Meena Pillai	meena.p@college.edu	1	88000.00
3	Dr. Sunil Rajan	sunil.r@college.edu	2	82000.00

23. Find all students whose email ends with '@college.edu' using the LIKE operator.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query: `select * from students where email like '%@college.edu';`. The 'Data Output' tab is active, displaying a table with 10 rows and 7 columns: student_id, first_name, last_name, email, date_of_birth, department_id, and enrollment_year.

student_id [PK] integer	first_name character varying (50)	last_name character varying (50)	email character varying (100)	date_of_birth date	department_id integer	enrollment_year integer
1	Arjun	Mehta	arjun.mehta@college.edu	2003-04-12	1	2022
2	Priya	Suresh	priya.suresh@college.edu	2003-07-25	1	2022
3	Rohan	Verma	rohan.verma@college.e...	2002-11-08	2	2021
4	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	Kavya	Menon	kavya.menon@college.e...	2002-05-17	2	2021
7	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023
8	Deepika	Rao	deepika.rao@college.edu	2003-08-09		
9						
10	rahul	sharma	rahul.sharma@college.e...	2003-05-10		

Successfully run. Total query runtime: 209 msec. 10 rows affected.

24. Count the total number of students per enrollment_year.

pgAdmin 4

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No limit

Query Query History

```
1 select enrollment_year,count(*) as total_students from students group by enrollment_year;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

enrollment_year	total_students
2022	5
2023	3
2021	2

Task 3: Multi-Table Joins

25. List each student's full name (first_name + ' ' + last_name) alongside the name of their department. (JOIN students and departments.)

pgAdmin 4

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college_db/postgres@PostgreSQL 18

No limit

Query Query History

```
1 select s.first_name || ' ' || s.last_name as student_name, d.dept_name from students s join departments d
2 on s.department_id=d.department_id;
```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

student_name	dept_name
Arjun Mehta	Computer Science
Priya Suresh	Computer Science
Rohan Verma	Electronics
Sneha Patel	Mechanical
Vikram Das	Computer Science
Kavya Menon	Electronics
Aditya Singh	Civil
Deepika Rao	Computer Science
rahul sharma	Computer Science

Successfully run. Total query runtime: 210 msec. 10 rows affected

Total rows: 10 Query complete 00:00:00.210 CRLF Ln 2,0

26. Show each enrollment along with the student's name and the course name. (3-table JOIN: enrollments, students, courses.)

The screenshot shows the pgAdmin 4 interface with a query window open. The query is a 3-table JOIN between enrollments, students, and courses. The results are displayed in a table with 10 rows.

```
1 select
2   e.enrollment_id,
3   s.first_name || ' ' || s.last_name as student_name,
4   c.course_name,
5   e.grade
6 from enrollments e
7 join students s
8 on e.student_id=s.student_id
9 join courses c
10 on e.course_id=c.course_id;
```

enrollment_id	student_name	course_name	grade
1	Arjun Mehta	Data Structures & Algorith...	A
2	Arjun Mehta	Database Management Sy...	B
3	Priya Suresh	Data Structures & Algorith...	B
4	Priya Suresh	Object Oriented Program...	A
5	Rohan Verma	Circuit Theory	A
6	Vikram Das	Database Management Sy...	A
7	Kavya Menon	Circuit Theory	B
8	Deepika Rao	Data Structures & Algorith...	A
9	Deenika Rao	Object Oriented Prooram...	R
10			

Successfully run. Total query runtime: 155 msec. 10 rows affected.

27. Find all students who are NOT enrolled in any course using a LEFT JOIN and WHERE ... IS NULL pattern.

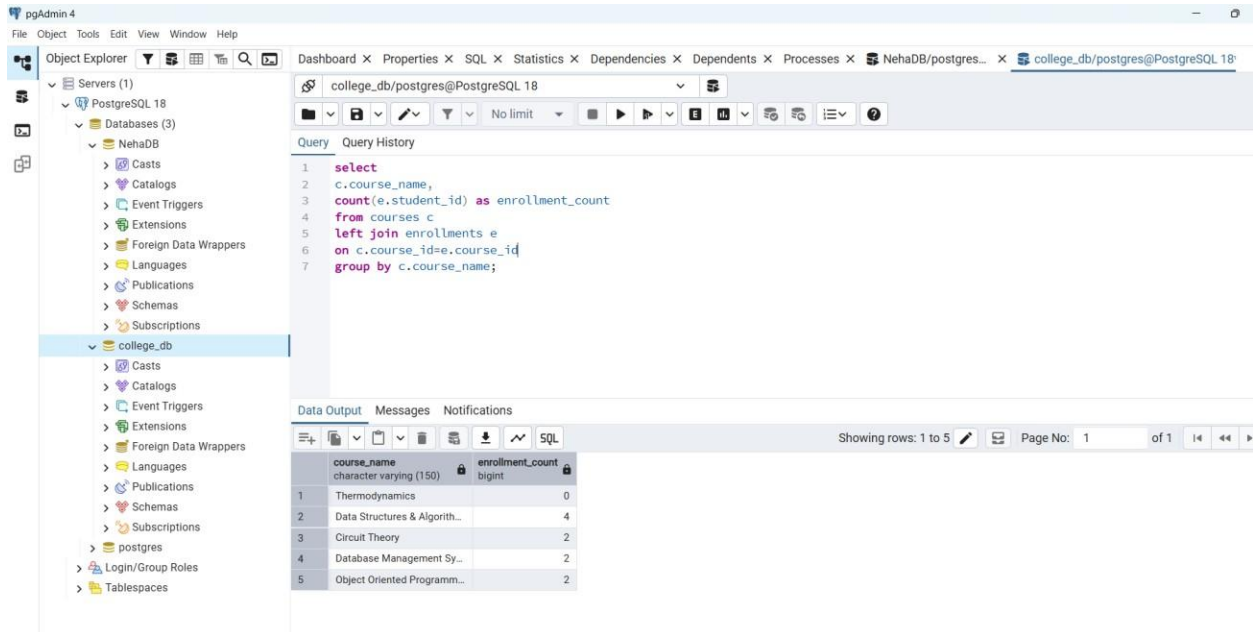
The screenshot shows the pgAdmin 4 interface with a query window open. The query is a LEFT JOIN between students and enrollments, filtering for students who are not enrolled in any course. The results are displayed in a table with 4 rows.

```
1 select s.student_id, s.first_name, s.last_name from students s left join enrollments e on s.student_id=e.student_id
2 where e.student_id is null;
```

student_id [PK]	first_name	last_name
11	anita	reddy
10	rahul	sharma
4	Sneha	Patel
7	Aditya	Singh

Successfully run. Total query runtime: 232 msec. 4 rows affected.

28. Display every course along with the number of students enrolled in it. Courses with zero enrolments must still appear. (LEFT JOIN courses with enrollments, GROUP BY course.)

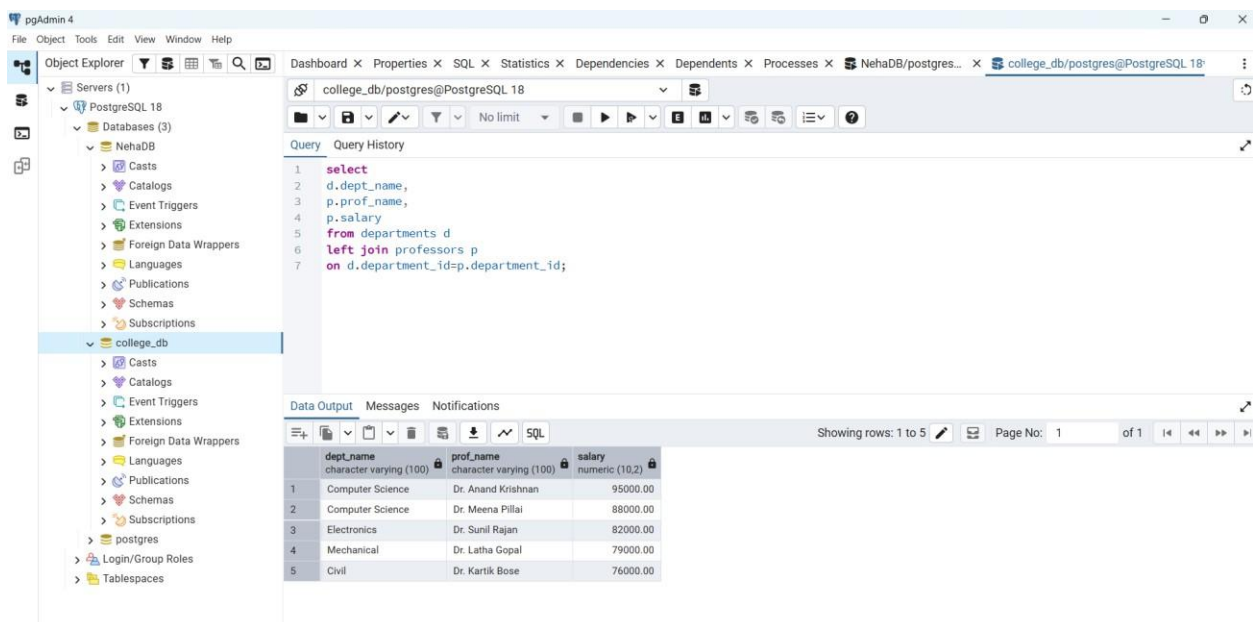


The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query that uses a LEFT JOIN between the 'courses' and 'enrollments' tables, grouped by course name. The 'Data Output' tab at the bottom displays the results of the query.

```
1 select
2 c.course_name,
3 count(e.student_id) as enrollment_count
4 from courses c
5 left join enrollments e
6 on c.course_id=e.course_id
7 group by c.course_name;
```

course_name	enrollment_count
Thermodynamics	0
Data Structures & Algorith...	4
Circuit Theory	2
Database Management Sy...	2
Object Oriented Programm...	2

29. List each department along with its professors and their salaries. Include departments that have no professors yet.



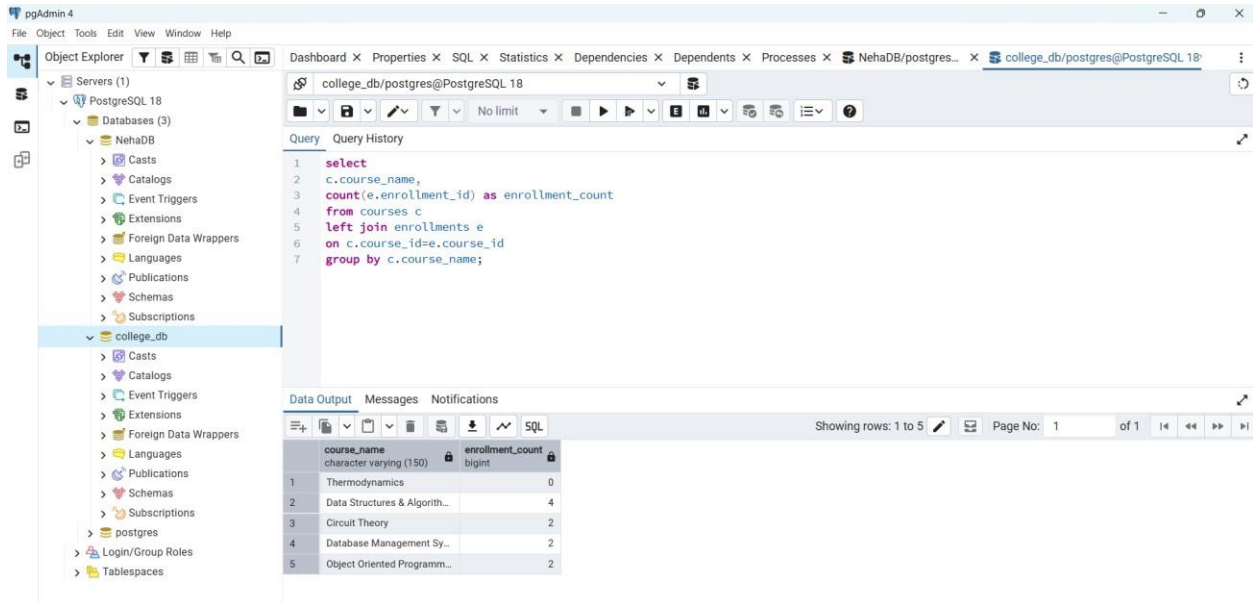
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query that uses a LEFT JOIN between the 'departments' and 'professors' tables. The 'Data Output' tab at the bottom displays the results of the query.

```
1 select
2 d.dept_name,
3 p.prof_name,
4 p.salary
5 from departments d
6 left join professors p
7 on d.department_id=p.department_id;
```

dept_name	prof_name	salary
Computer Science	Dr. Anand Krishnan	95000.00
Computer Science	Dr. Meena Pillai	88000.00
Electronics	Dr. Sunil Rajan	82000.00
Mechanical	Dr. Latha Gopal	79000.00
Civil	Dr. Kartik Bose	76000.00

TASK 4:

30. Calculate the total number of enrollments per course. Display course_name and enrollment_count.



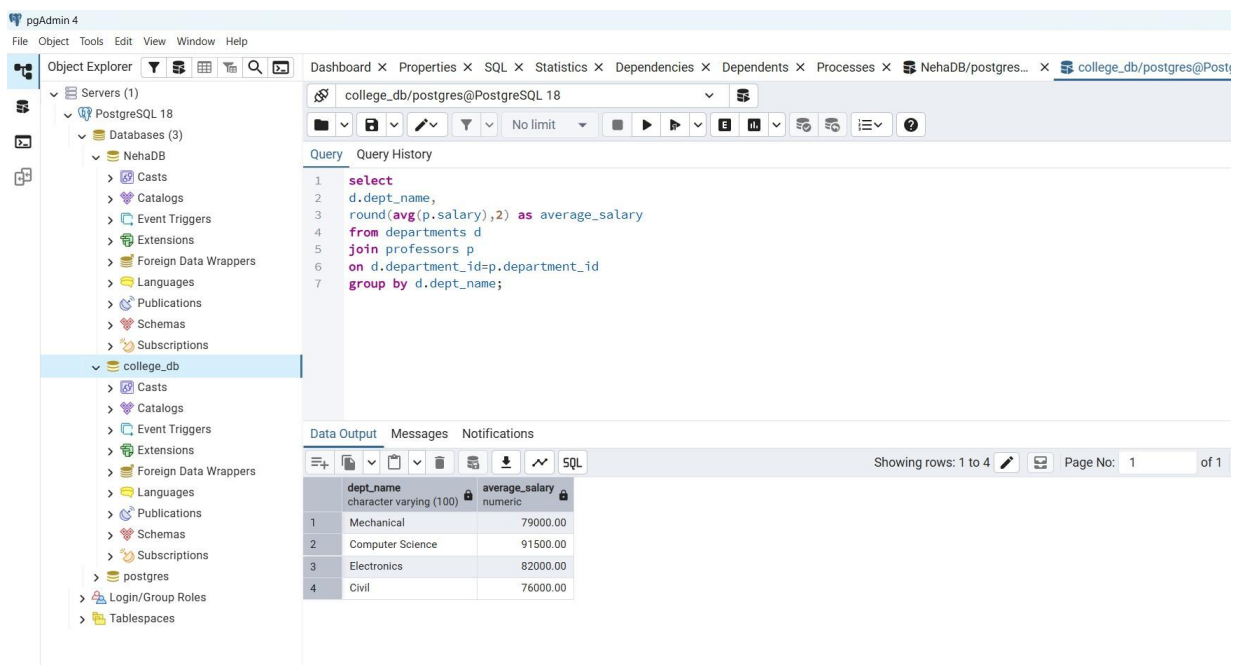
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query window with the following query:

```
1 select
2 c.course_name,
3 count(e.enrollment_id) as enrollment_count
4 from courses c
5 left join enrollments e
6 on c.course_id=e.course_id
7 group by c.course_name;
```

The 'Data Output' tab at the bottom shows the results of the query:

course_name	enrollment_count
Thermodynamics	0
Data Structures & Algorith...	4
Circuit Theory	2
Database Management Sy...	2
Object Oriented Programm...	2

31. Find the average salary of professors per department. Round to 2 decimal places.



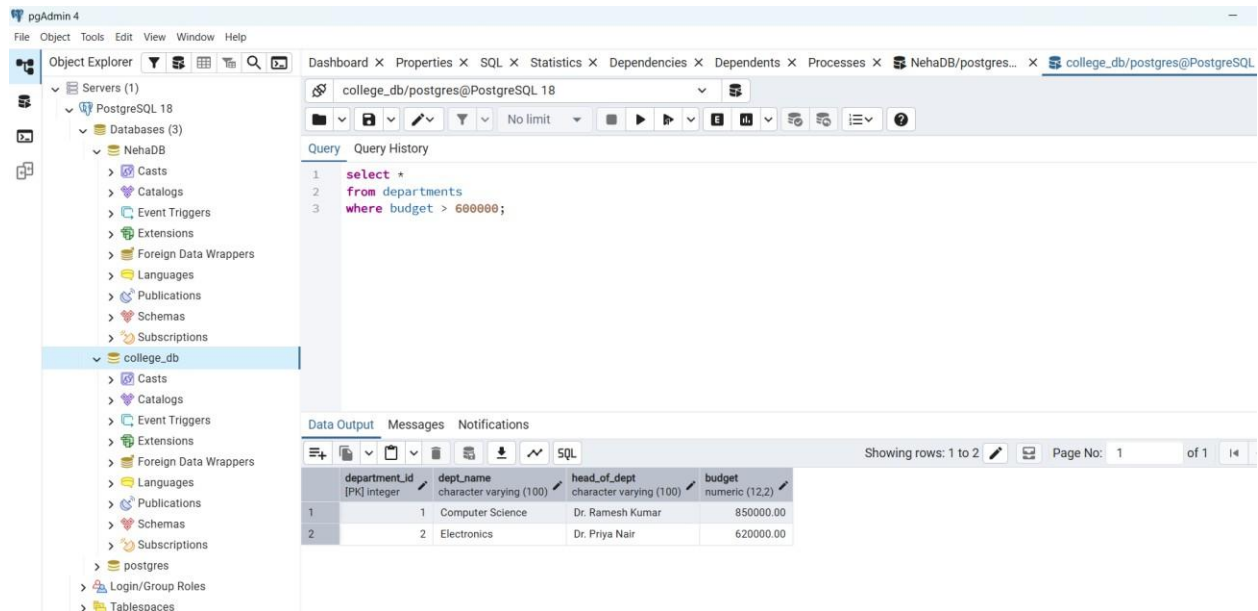
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query window with the following query:

```
1 select
2 d.dept_name,
3 round(avg(p.salary),2) as average_salary
4 from departments d
5 join professors p
6 on d.department_id=p.department_id
7 group by d.dept_name;
```

The 'Data Output' tab at the bottom shows the results of the query:

dept_name	average_salary
Mechanical	79000.00
Computer Science	91500.00
Electronics	82000.00
Civil	76000.00

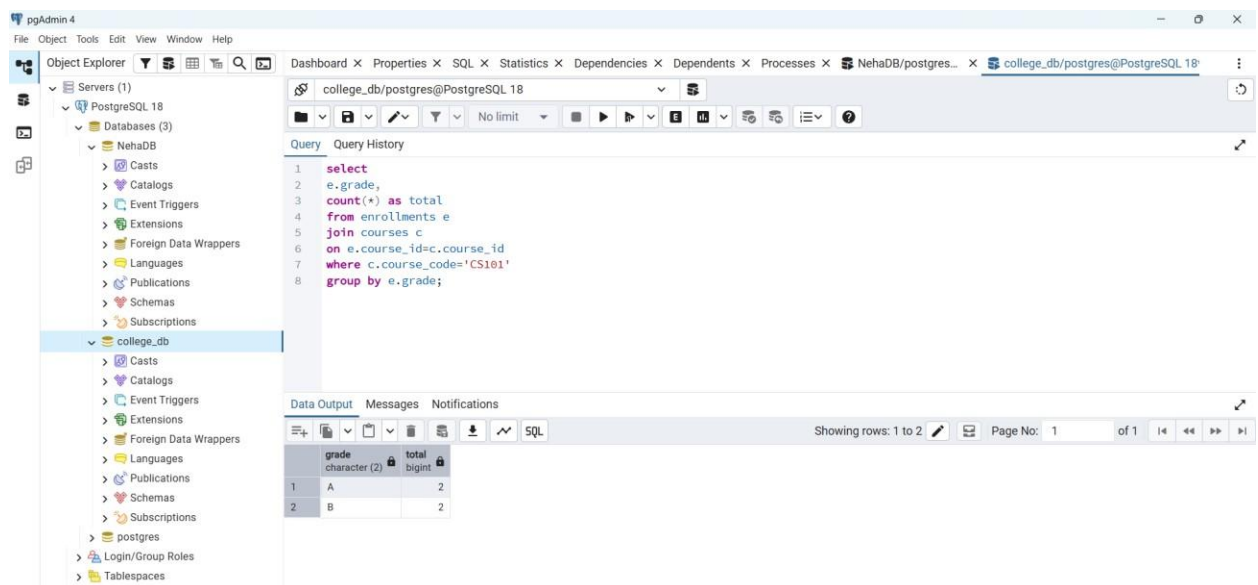
32. Find all departments where the total budget exceeds 600,000.



The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with 'college_db' selected. The main pane shows a SQL query: `select * from departments where budget > 600000;`. The 'Data Output' tab at the bottom shows the results of the query in a table with 4 columns: department_id, dept_name, head_of_dept, and budget. The results show two rows: Computer Science with a budget of 850,000.00 and Electronics with a budget of 620,000.00.

department_id	dept_name	head_of_dept	budget
1	Computer Science	Dr. Ramesh Kumar	850000.00
2	Electronics	Dr. Priya Nair	620000.00

33. Show the grade distribution for course CS101: count of each grade (A, B, C, D, F).



The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with 'college_db' selected. The main pane shows a SQL query: `select e.grade, count(*) as total from enrollments e join courses c on e.course_id=c.course_id where c.course_code='CS101' group by e.grade;`. The 'Data Output' tab at the bottom shows the results of the query in a table with 2 columns: grade and total. The results show two rows: A with a total of 2 and B with a total of 2.

grade	total
A	2
B	2

34. Using HAVING, list departments where more than 2 students are enrolled across all courses in that department.

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Query Query History

```
1 select
2 d.dept_name,
3 count(s.student_id) as total_students
4 from departments d
5 join students s
6 on d.department_id=s.department_id
7 group by d.dept_name
8 having count(s.student_id) > 2;
```

Data Output Messages Notifications

Showing rows: 1 to 2 of 1 Page No: 1

	dept_name character varying (100)	total_students bigint
1	Computer Science	5
2	Electronics	3